AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 06

WEEK OF 3–9 OCTOBER 2026

While It Waits

A run that has asked a person and not yet heard back is neither working nor done. This week five frameworks changed what that state reports.

40 releases read

5 mechanisms executed

10 papers read

14 models priced

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

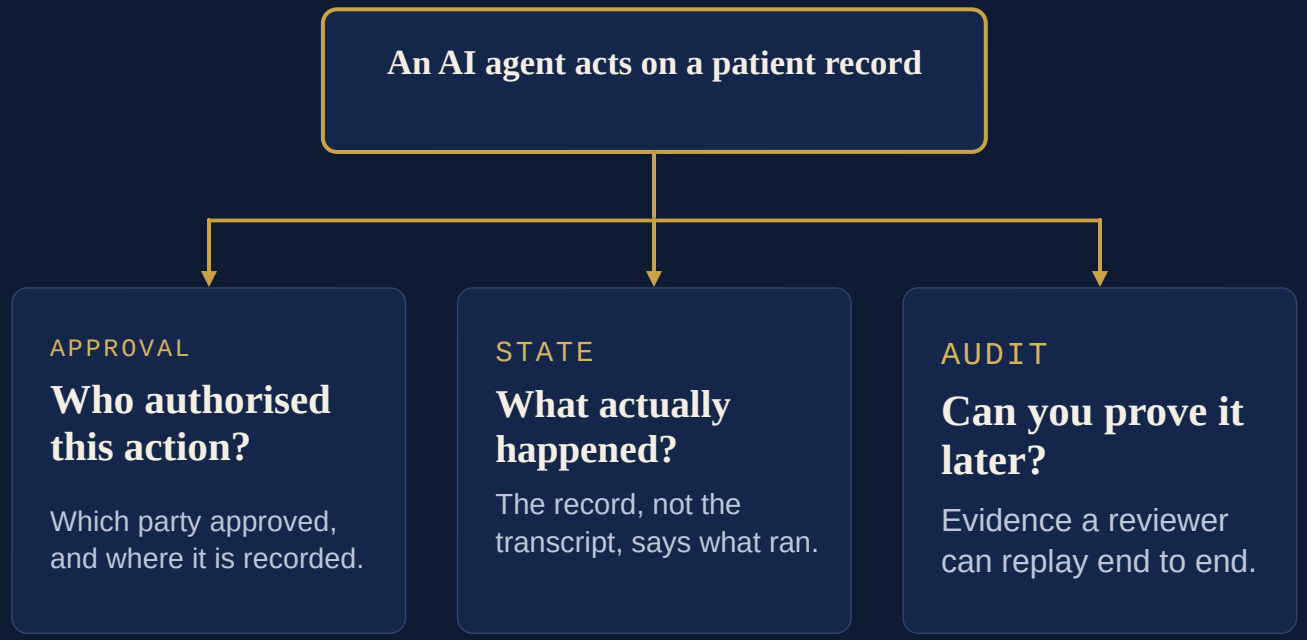
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

From the travelling yes to the waiting run

CONTEXT

Last week: the yes travelled

Issue 05 followed an approval to the agent it belongs to and the store it waits in. This week: the wait itself.

→

PROBLEM

The pause itself broke

An answered question stayed open, one pharmacist's approval became three signatures, and a hold was written before its question was answered.

→

SOLUTION

Own the inbox and the key

Track asked and answered interrupt ids in your own table, refuse a second answer, and give every gate an idempotency key.

STATE · 04–05
The answered question

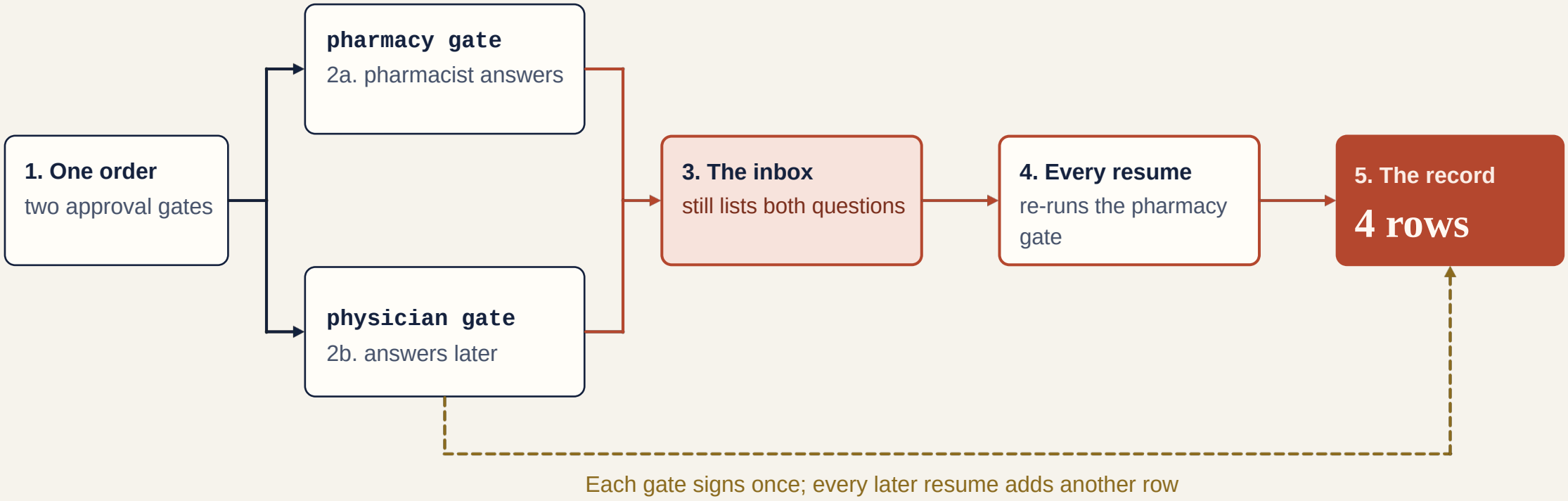
PAUSE · 06
What runs while it waits

REPORT · 07–08
What the caller is told

COST, DATES · 09–10
Prices and deadlines

EXECUTED MECHANISM · PROBLEM · LANGGRAPH 1.2.12

The answered question that stayed open



EXECUTED MECHANISM · SOLUTION · 1.2.13 AND YOUR HARNESS

Own the inbox, key every gate

STEP 1

Record asked and answered ids

In your own table, not the thread's list.

→

STEP 2

Refuse a second answer

Both versions still accept one silently.

→

STEP 3

Key every gate

An idempotency key on each signature write.

→

STEP 4

Drain, then upgrade

Paused threads keep the old rules.

4 to 2

signature rows for 2 approvals, 1.2.12 to 1.2.13

2 to 1

inbox items after one of two answers

3 rows

when a thread paused before the upgrade finished after it

> 1.2.13 (PR 9103) fixes how a finished gate is read. Checkpoints stored before the upgrade are not rewritten.

Three things happen while a run waits

CONTEXT
how it runs

Model emits calls → Each call is stamped → Paused for the clinician → Run resumes → Tool executes

PROBLEM
where it broke

GOOGLE ADK 2.11
Workflow-node tools skip the stamp check
DOCS

MS AGENT FW 1.21
The hold was written before the answer
EXECUTED

OPENAI JS 0.20
The tool ran, then the resume threw
EXECUTED

SOLUTION
what to build

Ordering is not consent

Never let a write share a response with the question it depends on. Pass a session on every path, reconcile on turn-budget errors, and store the stamp in the approval row.

The report and the record came apart

CONTEXT
how it runs

Agent calls a tool → Tool writes the row → Hooks write the audit → Context is summarised → Caller is told

PROBLEM
where it broke

FASTMCP 4.1.0
Timeout reported; the order was written
EXECUTED

PYDANTIC-AI 2.54
A cache hit ran no audit hooks
EXECUTED

LANGCHAIN 1.4.4
Middle omitted, logged only at DEBUG
DOCS

SOLUTION
what to build

Read errors as outcome unknown

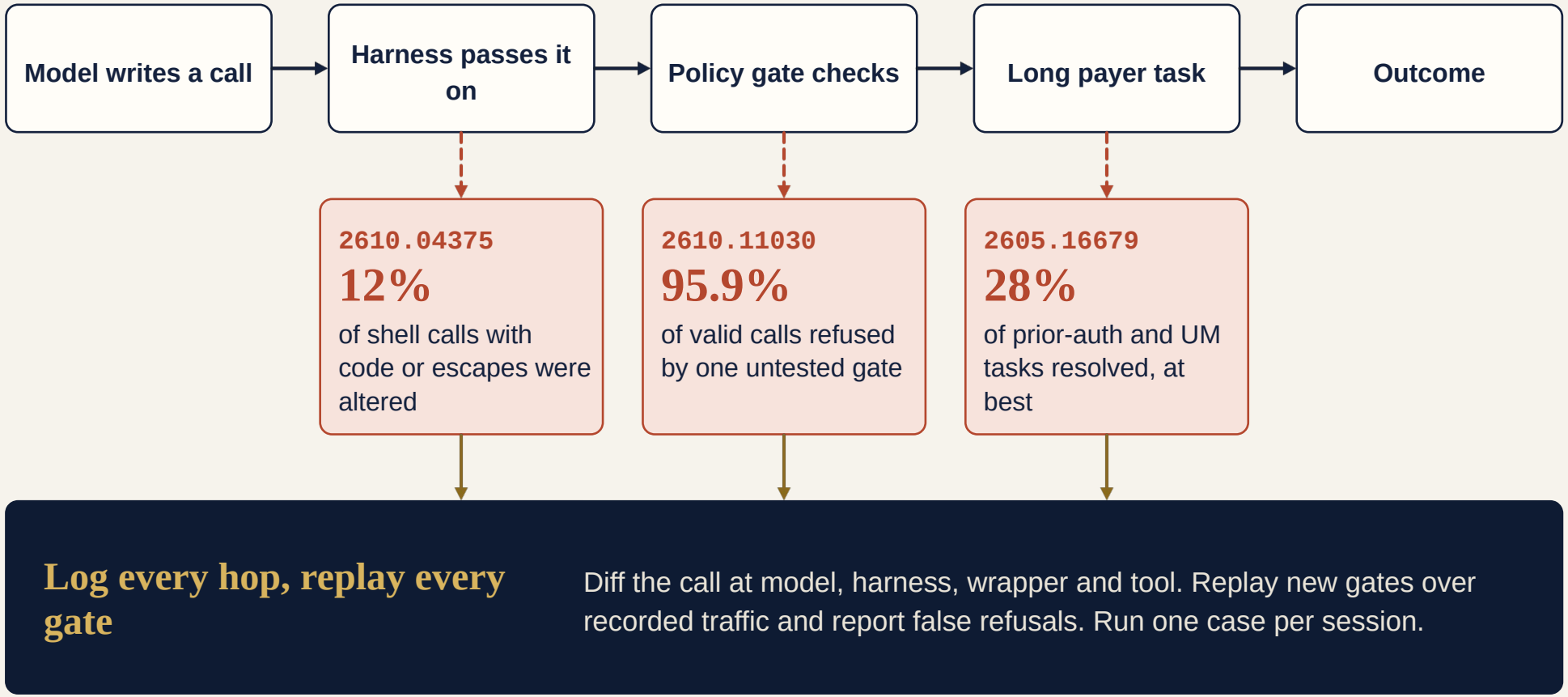
Look up a timed-out write by idempotency key before any retry, put mandatory audit in the outermost wrapper, and record when a summary fallback fired.

The call that ran was not the call written

CONTEXT
how it runs

PROBLEM
where it broke

SOLUTION
what to build



COST · LIST PRICES READ 9 OCTOBER

A price cliff at 100,000 tokens

MODEL	COST PER 10,000 DOCS	INDEX
Claude Haiku 5.5	$6.50; $32.50 over 100k	43
GPT-6 Luna	$6.50	38
Claude Sonnet 5.5	$130	56
GPT-6.1 Sol	$130	52
Kimi K3	$195	44
Claude Opus 5.5	$260	58

Per million tokens in/out: Haiku 5.5 and Luna $0.10/$0.50 (Haiku $0.50/$2.50 on any prompt over 100k), Sonnet 5.5 and 6.1 Sol $2/$10, Kimi K3 $3/$15, Opus 5.5 $4/$20. 4,000 in and 500 out per document, uncached. Index = Artificial Analysis Intelligence Index; its healthcare page did not load this week.

CONTEXT

Haiku 5.5, new on 7 Oct, does this workload for $6.50.

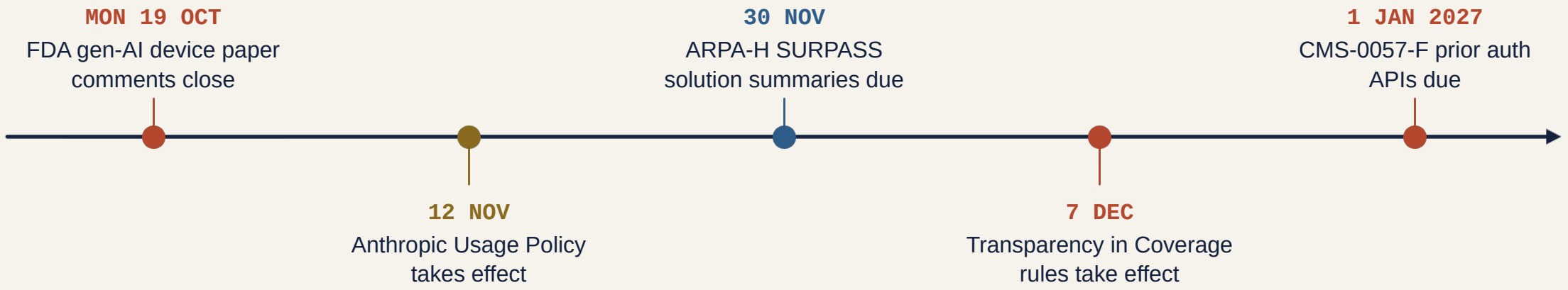
PROBLEM

One prompt over 100,000 tokens prices the whole request at 5x, cache hits included.

DECISION

Measure your prompt-length distribution before budgeting, and route long chart reviews deliberately.

Dates for your change-control file



- **REGULATORY**
- **VENDOR POLICY**
- **FUNDING**

Every framework upgrade before 1 Jan changes a regulated interface.

Funders now write ‘gated and verifiable’

ASTP/ONC · MEDSTAR

Gated agent middleware funded

A LEAP award for durable processes, patient-gated data pods and tamper-evident routing.

OPENAI

BAA in product, Decisions beta

In-product BAA and a HIPAA setting (5 Oct); Decisions API beta on gpt-6-luna (6 Oct).

ANTHROPIC

A human stays in the loop

Usage Policy from 12 Nov keeps a qualified human and AI disclosure for health uses.

ANACONDA

MCP server at GA

Remote, read-only, four tools. Remove the beta local package.

HL7 · ARPA-H

MCP for HIE; an agentic ops RFP

An HIE MCP server among HL7 AI Challenge winners; SURPASS asks for an agentic operations layer.

QUIET THIS WEEK

Epic, FDA on AI, OCR, the MCP specification, Moonshot, NVIDIA. Nebius bought Inferize with no user-facing change.

Seven moves, with what forces each

Move	Why	By
Build the inbox from your own asked and answered interrupt ids	Answered questions stayed open	Before upgrade
Key every gate; drain paused threads first	Old checkpoints keep old rules	Before upgrade
Keep dependent writes out of the question’s response	Writes ran before the answer	Now
Treat write-tool timeouts as outcome unknown	The error came after the row	Now
Move mandatory audit to the outermost wrapper	Cache hits skip inner hooks	Next release
Measure prompt lengths before budgeting Haiku 5.5	5x price over 100k tokens	Before budget
Freeze framework versions behind prior auth APIs	CMS-0057-F APIs due	1 Jan 2027

AV AI VELOCITY

THIS WEEK’S POSITION

Ordering is not consent. Only your gate records the decision.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
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github.com/zkumar/AI-Velocity-Technology-Update-